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Review article

PPG-based biometric authentication: A review on architectures, datasets, attacks and security challenges

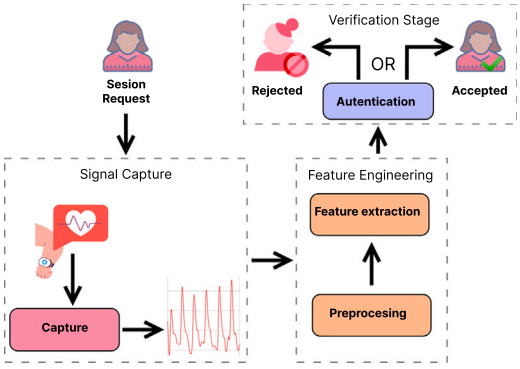
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HIGHLIGHTS

- Systematic analysis reveals a major security gap in PPG authentication literature.
- In-depth analysis of key threats, including spoofing, replay, and presentation attacks.
- Advocates for a paradigm shift from a classification to a security-first approach.

GRAPHICAL ABSTRACT



ARTICLE INFO

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Photoplethysmography (PPG)
Biometric authentication
Continuous authentication
Wearable devices
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ABSTRACT

Recent Photoplethysmography (PPG-based) authentication approaches are emerging as promising biometric modalities, driven by the widespread adoption of wearable devices capable of noninvasive and continuous identity verification. These advances have also introduced new challenges and threats to security. This review presents a taxonomy of PPG-based authentication and identification systems, emphasizing a comparative analysis of different authentication models using PPG signals, in terms of performance, preprocessing algorithms, sensors and hardware platforms, as well as security. The lack of standardized datasets, which limits reproducibility and generalization across demographics and real-world conditions, is discussed. Unlike other studies, this work focuses on the impact and mitigation strategies of the main attacks, such as spoofing, replay, and presentation, in PPG-based authentication systems. Additionally, based on the gaps and challenges identified, a security analysis is presented, emphasizing standardized metrics such as False Acceptance Rate (FAR), False Rejection Rate (FRR), and Equal Error Rate (EER), and a dataset standardization for scalability. Finally, this research highlights new challenges and future work, such as the integration of robust ML architectures with standardized datasets and security evaluations to enable reliable PPG-based authentication systems in real-world applications.

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1. Introduction

In recent years, security breaches in credentials and growing interest in abandoning this authentication factor [7,9,34,36] have driven the search for more secure and flexible identity verification methods. As a result, biometrics have captured the attention of the research community [32,46,67], which has made significant advances in the area. Although biometrics can mitigate certain vulnerabilities associated with credentials, they also present significant threats [35,44,54,69,78], especially so-called static biometrics (fingerprints, facial recognition, iris recognition, among others). In addition, many of these techniques require the active participation of the user in the authentication process, which limits their applicability in advanced modalities such as continuous authentication.

In this context, physiological signals acquired by wearable devices are a promising avenue, as they are considered passive biometrics and require minimal user involvement for capture. Photoplethysmography (PPG) is a noninvasive and dynamic method for detecting variations in blood volume, making it a suitable option for systems that require continuous identity verification [59]. This signal has generated growing interest in the scientific community, leading to numerous studies on PPG-based authentication. Despite advances, a gap remains: the existing literature is scattered and lacks a comprehensive, security-focused analysis that fully articulates authentication systems, their vulnerabilities, the limitations of datasets, and the evaluation methods needed to guide future development in the field.

Fig. 1 illustrates the typical workflow of a PPG-based authentication system. It begins with a session request from the user and proceeds through three main components: (1) PPG signal acquisition using wearable devices (usually placed on the wrist); (2) a feature-engineering pipeline that pre-processes the raw signal (filtering, motion artifact removal, segmentation, etc.) and extracts discriminative features for biometric template generation; and (3) the verification stage, where decision strategies such as machine learning or deep learning classifiers, distance-based measures, or adaptive thresholds determine whether to accept or reject the session. Fig. 1 summarizes the recurring technical blocks found in the reviewed works and highlights where potential security risks may exist.

The main contributions of this article are listed below:

- Taxonomy based on authentication and identification systems that use the PPG signal.
- Systematic and comparative review.
- Proposed evaluation metrics.
- Dataset guidance.
- Review of attacks.
- Identification of persistent challenges.

The rest of the article is organized as follows: Section 2 summarizes related work and highlights the differences in our approach. Section 3 describes the methodology used for article selection. Section 4 presents a taxonomy and analysis of PPG-based authentication models. Section 5 reviews the attack vectors relevant to these systems. Section 6 provides recommendations for evaluating authentication systems. Finally, Sections 7 and 8 discuss the general implications and future work of the research, and present the conclusions, respectively.

2. Related work

Although several previous reviews have addressed PPG-based authentication, there remains a lack of a critical integration of authentication architectures, attack vectors, and inherent security challenges, as shown in Table 1, which uses the notation ✔ (**Implemented**), ◗ (**Partially implemented**), and ✘ (**Not implemented**), summarises the differences between the scope of this work and that of previous relevant reviews, highlighting the strengths of our contribution.

Some studies have focused on PPG signal processing for specific purposes. Biswas et al. [5] reviewed techniques for estimating heart rate from wrist-based PPG signals and included a list of datasets relevant to this task. Although their work is relevant, it only briefly mentions the potential of PPG for biometric identification. Although their main focus is not on authentication systems or security aspects, their work is useful because it includes components of authentication systems. In the field of cardiac biometrics, Rathore et al. [58] examined Heart Biometrics, describing the functioning of the PPG signal, the types of sensors, the feature extraction methods, and the classifiers used. Although their discussion mentions privacy and security aspects, the treatment

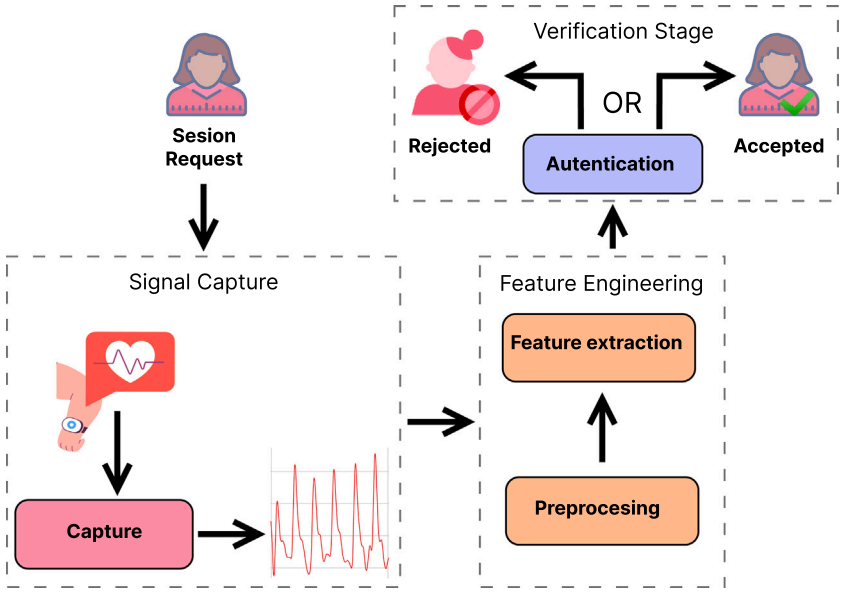


Fig. 1. Typical workflow of PPG-based authentication system showing (1) session request, a user requests authentication signal acquisition from wearable devices, (2) feature engineering pipeline with preprocessing and extraction stages and, (3) authentication handling with accept/reject outcomes.

Table 1
Comparison of related works with our proposed study.

Criterion	[5]	[58]	[62]	[15]	[42]	[41]	[6]	Ours
Systematic review methodology	✘	◗	◗	◗	◗	✔	✘	✔
Authentication-Based Taxonomy	✘	✘	✘	✘	✘	✘	✘	✔
Analysis of challenges in PPG	◗	◗	✔	✔	◗	✔	✘	✔
Focus on continuous authentication	✘	◗	✘	✔	✘	✔	✔	✔
Analysis of the potential of behavioral biometrics	✘	✘	✘	✘	✘	✘	✘	✔
Analysis of authentication architectures	✘	◗	◗	✔	✘	◗	✔	✔
Analysis of hardware used for PPG signal acquisition	✘	✘	✘	✘	✘	✘	✘	✔
Security and cancellability discussion	✘	✘	✘	✘	✘	◗	✘	✔
Proposed indicators for authentication systems	✘	✘	✘	✘	✘	✘	✘	✔
Limitations and future challenges	◗	◗	◗	✔	◗	✔	◗	✔

of continuous authentication and multimodal schemes was limited and did not detail the implications of the architectures or the possible vulnerabilities of PPG data.

On the other hand, Shahid et al. [62] analyzed the use of ECG (Electrocardiogram), PPG, and PCG (Phonocardiography) in biometric systems. In the specific case of the PPG signal, they evaluated preprocessing strategies to mitigate motion artifacts, feature extraction techniques and the use of machine learning algorithms. They also highlighted the need for larger and more representative datasets to improve robustness. Although their review does not delve exhaustively into the vulnerabilities associated with signal acquisition, it is a comprehensive work that summarizes various relevant aspects of authentication with PPG signals.

Continuous authentication based on PPG has been the focus of more specialized reviews. For example, Donida Labati et al. [15] classified preprocessing methods and distinguished between PPG and R-PPG, comparing public datasets in the context of continuous authentication. However, their analysis did not comprehensively address the data generation conditions relevant to adversarial scenarios, nor the security challenges inherent in continuous monitoring. Similarly, Boshoff and Hancke [6] reviewed continuous authentication methodologies with PPG, emphasizing architectural proposals and performance indicators, but with a limited and superficial exploration of specific attacks and countermeasures.

More recently, Li et al. [41] proposed a four-layer model for PPG-based biometric authentication (Acquisition, Conditioning, Representation, and Authentication). This work examines techniques by layer and classifies authentication methods. The review highlights vulnerabilities and suggests evaluation criteria that include cancelability, transparency, continuity, and accessibility; it also addresses threats such as template theft and adversarial attacks on AI models. However, it does not provide a systematic exploration of attack categories, their feasibility under different system designs, or specific mitigation strategies. Finally, Lien and Vhaduri [42] addressed more specific challenges of PPG signals. In their work, they mention biometric degradation and difficulties in IoT environments; however, these issues are often presented as isolated cases or in limited-scale studies, without connecting them to a broader analysis of architectures, datasets, or threat landscapes.

Taken together, existing reviews provide valuable insights into PPG signal processing and its applications in biometrics and continuous authentication. However, there remains a need for an integrated and critical treatment that simultaneously considers system architectures, plausible attack vectors and associated security implications; this gap is the central objective of the present review.

3. Methodology

Based on the identified need to overcome the limitations of previous reviews, we conducted a systematic review of the literature following the PRISMA guidelines [52], which guarantee methodological rigor. The search strategy was designed to be exhaustive and precise, based on three key components: the selection of relevant keywords, an iterative refinement of the criteria, and multi-stage filtering.

3.1. Search strategy

In order to find the most relevant articles, it is essential to refine the search criteria. For this task, a series of keywords (K) were defined, which were combined with each other to ensure search relevance, always respecting the coherence of the combinations in the context of biometric authentication.

To guarantee the appropriate combination of keywords, we used a set of keywords K_{coh} which are used as search criteria. The definition of this subset is based on:

$$K_{\text{coh}} = \left\{ S \in \mathcal{P}(K) \mid |S| > 1,\ S \cap P \neq \emptyset,\ C(S) = 1 \right\} \tag{1}$$

where P contains all synonyms of PPG and similar words, and C is defined as:

$$C(S) = \begin{cases} 1, & \text{if } \text{sim}(S, \text{"biometric authentication"}) > \theta, \\ 0, & \text{otherwise}, \end{cases} \tag{2}$$

This approach combined controlled vocabulary (e.g., `PPG + "Continuous Authentication"`) with iterative refinement based on preliminary search results. The strategy intentionally included nonstandard terminology combinations to capture emerging research at the intersection of physiological sensing and security.

3.2. Selection criteria

Once the search criteria K_{coh} were defined, they were applied to the seven databases used in this work: IEEE Xplore, ScienceDirect, Scopus, ResearchGate, MDPI, ACM Digital Library and Springer Link.

Articles were subsequently screened based on title, abstract, and full text, applying strict inclusion and exclusion criteria.

Inclusion criteria ensured that studies advanced PPG-based authentication, including original peer-reviewed research published in journals or conferences, and focused on system architectures, methods, datasets, or security threats and vulnerabilities.

Exclusion criteria maintained the integrity of the review by omitting studies outside the selected timeframe, irrelevant to authentication, lacking empirical findings, not peer-reviewed, lacking performance metrics, providing insufficient methodological details, or having inaccessible full texts.

This rigorous methodological approach ensures comprehensive coverage of the current state-of-the-art in PPG-based authentication while maintaining the quality standards necessary for a systematic review. The selected works span from 2022 to May 2025, a period in which we find the most relevant work in the field, particularly the proliferation of deep learning architectures and the growing emphasis on security analysis in response to the increasing deployment of wearable devices for continuous authentication. This curated collection of 32 articles forms the foundation for the comparative analysis and critical discussion in the subsequent sections. It enables us to identify both cutting-edge technological advances and persistent research gaps that require further investigation.

4. PPG-based authentication models

The selected PPG-based authentication studies cover a wide range of approaches. To analyze these studies systematically, we propose a taxonomy that classifies them into two main categories: user identification and

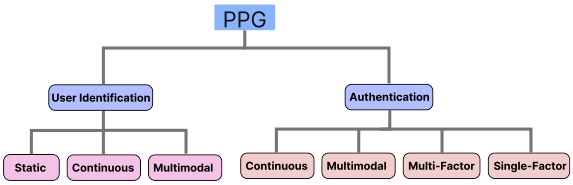


Fig. 2. Taxonomy of PPG-based authentication and identification systems.

authentication. Although this work focuses solely on authentication, we consider it relevant to include identification because many techniques can also be applied to authentication processes. Under user identification, works are divided into static (one-time identity verification) and continuous (ongoing identity monitoring) strategies. For authentication, the proposed taxonomy distinguishes between multimodal (integration of PPG with other signal or data), multifactor (combination of PPG with non-biometric factors such as credentials), and single-factor (exclusive reliance on PPG signals). This classification framework facilitates a systematic analysis of design choices, technological implementations, and performance tradeoffs across the surveyed literature. Fig. 2 illustrates the proposed taxonomy.

This review is structured into two main tables. The tables included serve a dual purpose: to summarise the reviewed literature and to provide a comparative analysis that allows for the rapid identification of trends, strengths, and limitations of each approach. To achieve this, they were organized into two complementary levels of analysis, one focused on their functional and safety properties, and the other on the technical performance of the models.

Table 2 includes fields that focus on functional and security properties. Body location, artifact mitigation, and feature extraction enable the assessment of signal quality and the model's discriminatory capacity. Behavioral analysis, continuous authentication, and multimodal integration illustrate the practical scope of each approach, while cancellability and attack evaluation reflect the robustness of the systems in real-world usage and threat scenarios.

Table 3 focuses on technical performance. Preprocessing algorithms are included due to their direct influence on PPG signal quality, while classifiers allow us to identify methodological developments and their relationship with the results obtained. Cross-validation is presented as an indicator of statistical robustness, and evaluation metrics provide a standardized basis for objective comparison between studies. Among these, accuracy stands out as the main metric, as it is the most commonly reported in most studies and offers a quick reference for comparing the performance of different approaches.

The notation employed in these two tables is as follows: ◗ (**Partially**), ✔ (**Implemented**), and ✘ (**Not implemented**).

The following subsections synthesize recent works grouped according to this classification, beginning with user identification and then detailing the authentication categories: single-factor, multimodal, multifactor, and continuous. Within each subcategory, techniques based on behavioral biometrics are also included, given the growing importance of this area and its integration into various authentication strategies.

4.1. User identification

Identification is based on distinguishing between known and unknown users. Unlike authentication, where the goal is to accept or reject a user's access, identification does not restrict access to a resource. The close relationship between identification and authentication makes the methods presented in identification studies relevant for authentication tasks, as they introduce techniques that could be applied in these contexts.

Modeling temporal dependencies in PPG-based biometric systems remains a central and unresolved challenge. Recent advances highlight inherent tradeoffs between methodological sophistication and practical applicability. For example, bidirectional LSTM architectures can effectively capture temporal dynamics; however, the perfect training accuracy (100 %) reported by Cherry et al. [11], despite L2 regularization, suggests potential overfitting, raising concerns about reliability. In contrast, Ngo et al. [50] mitigate such issues through an integrated hardware–software framework that enhances signal reliability using optical photodetectors (OPDs) and real-time quality assessment, albeit at the cost of increased system complexity and limited scalability. Similarly, the 2D Gram matrix approach proposed by Cherry et al. [12] enriches the feature space by encoding spatial-temporal correlations, achieving high accuracy but incurring computational costs that limit efficiency in real-world deployments.

Other deep architectures emphasize advanced feature extraction and fusion. The Dual-domain and Multiscale Fusion Deep Neural Network (DMFDNN) introduced by Liu et al. [45] employs a two-branch architecture with an attention mechanism to integrate discriminative features from time and frequency domains. While promising results are reported, including references to floating-point operations and parameter counts, the study lacks a thorough analysis of energy efficiency or inference time which are critical for deployment on portable devices or in real time applications. To enhance scalability for large datasets, Wei et al. [70] extract multiscale features using PPG-specific data augmentation and curriculum learning (combining cross-entropy and triplet losses), improving discrimination and robustness over manual methods or shallow networks. However, performance may suffer from augmentation-induced ambiguities and limited generalization across datasets.

Alternative methods and fusion strategies further aim to enhance robustness. Drawing on techniques from vocal recognition, Přibil et al. [57] extract features from the second derivative of the PPG waveform (SD-PPG), including temporal ratios and partial areas, and model them using Gaussian Mixture Models (GMMs) to evaluate accuracy under electromagnetic interference, such as in low-intensity MRI environments. Meanwhile, Hwang et al. [27] address temporal stability across sessions via score-level fusion: multiple inputs are generated using temporal stretching techniques (cubic, DTW, Fourier padding), and scores from stable models are combined to improve overall performance.

Collectively, these studies highlight a persistent tension in PPG modeling: approaches that maximize discriminative power often compromise adaptability, computational efficiency, or practical feasibility. Consequently, an optimal method should align with application-specific requirements, such as real-time processing, noise robustness, and resource constraints.

4.2. Single-factor authentication

Single-factor authentication (or one-factor authentication) verifies a user's identity using a single source of information. Despite growing interest in multimodal, multi-factor authentication (MFA), and continuous authentication methods in recent years, single-factor authentication still faces significant challenges. Recent studies have employed diverse machine learning and deep learning methodologies to enhance the accuracy of single-factor authentication using PPG.

One active area of research is the modeling of temporal characteristics using recurrent networks. Ben Salah et al. [3] devised a model of three consecutive LSTM layers to examine segmented frames of 50 points, distinguished by its computational efficiency achieved through preprocessing with a Butterworth filter and normalization, while maintaining good accuracy. In addition, Ortiz et al. [51] employ Bi-LSTM models directly on unprocessed PPG signals obtained from smartphones for user classification, eliminating the need for explicit filtering stages and showcasing the viability of this minimalist methodology. Another important avenue explores the use of convolutional networks (CNNs) and the transformation of 1D signals into 2D representations to extract spatial or spatio temporal patterns. Yang and Fang [71] use a CNN to generate multidimensional representations and apply Dynamic Weighted Triplet Loss, an innovation that

improves stability and scalability by eliminating the need for retraining when incorporating new users. Innovating in input representation, Zhao et al. [75] transform 1D PPG signals into 2D images using MTF (Markov Transition Field) to process them with a lightweight CNN (LW-CNN) optimized for wearable devices. Following this line of transformation to 2D, Ibrahim et al. [28] convert the signals into scalograms (using the continuous wavelet transform, CWT) and employ a hybrid architecture that combines a convolutional vision transformer (CVT) with ConvMixer, enhanced by attention layers to capture complex patterns and improve discrimination. Seok et al. [61] propose Siamese 1D-CNN networks that evaluate the similarity between two signals for authentication processes. Although the previous approaches present several advantages, the tendency to use deep learning is often poorly justified compared to other classical machine learning algorithms.

In the context of resource-constrained edge devices, such as those found in social vehicular ad hoc networks (SVAN) or the Internet of Things (IoT), Chen et al. [10] combine neural architecture search (NAS) with spiking recurrent neural networks (SRNN), achieving efficient authentication tailored to these platforms.

Table 2
Comparative Analysis of PPG-based Authentication Systems.

R	Datasets	Body Location	MA	Features	BA	CA	Combination	C	Attacks
[71]	KMU	Wrist	✘	CNN extraction	✘	✘	–	✘	–
[75]	BIDMC, Respiration Dataset	Fingertip,	✘	Converts PPG to 2D images, CNN extracts features	✘	✘	–	✘	–
[10]	Stress, WESAD	Wrist, Wrist and chest	✘	CWT, time FD	✘	✘	–	✘	–
[16]	BIDMC	Fingertip	✔	MB, SPP, self-attentio	✘	✘	ECG	✘	–
[33]	Self	Wrist	✔	EfficientNet-based Feature Extraction	✘	✘	Accelerometer, Gyroscope	✘	–
[8]	Self	Wrist	✔	FP, Multiple cardiac cycles	✘	✘	Credentials	✔	Random attack, Replay attack
[66]	Self	Wrist	✔	miniROCKET-based Feature Extraction	✔	✘	Credentials	✘	Random attack, Emulating attack
[79]	Self	Wrist	✔	Num Troughs, Num Crest, RA, Half Gesture Time, Trough Width, Gesture Duration, SA, CW, TD, FD, time FD	✔	✘	Accelerometer	✘	Random Attack, Brute Force Attack, Replay Attack, Synthesis Attack
[74]	Self	Wrist	✘	SA, RA, Press Width, Ratio Width, Troughs, Num Crest, Number of Trou ghs	✔	✘	Accelerometer	✘	Presentation Attacks, Random Attacks
[61]	Real-World PPG	–	✘	1D-CNN Feature Extraction	✘	✘	–	✘	–
[77]	Self, IEEE TBME 2013	Wrist, Fingertip	✔	SA, PW, RPA, CT, RA	✘	✔	–	✘	Random Attack, Synthesis Attack
[56]	IEEE TBME 2013, Real-World PPG, BioSec1	Fingertip, Fingertip, Fingertip	✔	Multiwavelet decomposition features	✘	✔	–	✘	–
[37]	Self	In-Ear	✔	SA, CT, PW, RPI, RA	✘	✔	–	✘	–
[48]	Self	Wrist	✔	TD, Benjamini-Yekutiel selection	✔	✘	–	✘	–
[43]	Self	Wrist	✘	Eigenhearts through HR windows	✘	✔	–	✘	–
[14]	CapnoBase IEEE TBME 2021, BIDMC, TROIKA	Fingertip, Fingertip, Wrist	✘	CNNs extract features	✘	✘	ECG	✘	–
[68]	BIDMC, MIMIC, IEEE TBME 2013	Fingertip, Fingertip, Fingertip	✘	TD, FD, Wavelet	✘	✔	–	✘	–
[28]	Real-World PPG, BIDMC, MIMIC	Fingertip, Fingertip, Fingertip	✔	Images from 2D scalograms	✘	✘	–	✘	–
[73]	Self, CapnoBase, MIMIC, BIDMC, Oxford	Face (with Camera), Fingertip, Fingertip, Fingertip, Remote (Camera)	✔	Statistical features, curve width, FD, FP	✘	✔	–	✔	–
[2]	BIDMC, Real-World PPG	Fingertip,	✘	TD, FD, IF, SE	✘	✔	ECG	✘	Spoofing Attack, Tampering Attack, Registration Phase Attack
[13]	Self	Wrist	✘	Raw PPG time series	✘	✔	–	✘	–
[63]	Pulse Transit Time PPG, BIDMC	Fingertip, Fingertip	◗	Mel-frequency cepstral coefficients (MFCCs)	✘	✘	ECG	✔	–
[53]	Self	Wrist	◗	heartbeat data, motor vibration, raw PPG	✔	✘	Motor vibration	✘	Random Attack, Replay Attack, Synthesis Attack
[51]	Self	Fingertip	◗	Raw PPG, FD, PPD, PTI, SR, TV, PPI, FHR, RMSSD, STD	✘	✘	–	✘	–
[3]	Real-World PPG	–	✔	Using three LSTM layers for feature extraction	✘	✘	–	✘	–

(*continued on next page*)

Table 2 *(continued)*

R	Datasets	Body Location	MA	Features	BA	CA	Combination	C	Attacks
User Identification									
[45]	BIDMC, MIMIC, IEEE TBME 2013, Biosec1	Fingertip, Fingertip, Fingertip, Fingertip	✔	TD, FD	✘	✘	–	✘	–
[57]	Self	Fingertip	✘	TF, Partial areas around peaks	✘	✘	–	✘	–
[50]	Self, MIMIC-II	Fingertip, Fingertip	◗	CNN extraction	✘	✘	–	✘	–
[11]	Real-World PPG	–	✘	BiLSTM extraction	✘	✘	–	✘	–
[70]	ZJPhys, CapnoBase, BIDMC, Biosec2	Fingertip, Fingertip, Fingertip, Fingertip	✘	PulseID DCNN Multi-scale Feature Extraction	✘	✘	–	✘	–
[12]	Self	Fingertip	◗	2D Gram matrices	✘	✘	–	✘	–
[27]	IEEE TBME 2013, TROIKA, Biosec1, Biosec2	Fingertip, Wrist, Fingertip, Fingertip	✔	Learned features from a CNN, derived from standardized and stretched PPG single pulses using Cubic spline interpolation, Dynamic Time Warping (DW), Padding wi th Fourier (PF), and Padding with Inverse Fourier (PIF)	✘	✘	–	✘	–

Note: MA: Motion Artifacts, BA: Behavioral Analysis, CA: Continuous Authentication, C: Cancelability. Abbreviations used: SA (Systolic Amplitude), PW (Pulse Width), CT (Crest Time), CW (Crest Width), RA (Ratio of Amplitude), RPI (Ratio of Pulse Interval), FP (Fiducial Points), AD (Angle-Domain), TD (Temporal Domain), MB (MultiRes Blocks), SPP (Spatial Pyramid Pooling), FD (Frequency Domain), PD (Power Domain), PPD (Peak-to-Peak Amplitude Difference Interval), PTI (Peak-to-Trough Time Interval), SR (Slope Ratio), TV (Trough Values), PPI (Peak-to-Peak Time Interval Domain), FHR (Frequency of Heart Rate), RMSSD (Root-Mean-Square of Successive Difference), STD (Standard Deviation).

Table 3
Comparative Performance Analysis of PPG-based Authentication Systems: Methodological Approaches, Accuracy and Other Metrics.

R	Preprocessing Algorithms	Classifiers or algorithms	CV	Accuracy	Others indicators
[71]	Low-pass filters, high-pass filters, DWT, SoCha	Dynamic Weighted Triplet Loss	✘	97.4 %	EER (2.6 %), FRR (4.6 %), FAR (1 %)
[75]	3rd-order Butterworth bandpass filter, amplitude normalization	LW-CNN	✘	96.17 %	Precision (96.24 %), recall (96.28 %), F1-score (96.11 %)
[45]	Fourth-order Butterworth filter, Variational mode decomposition	DMFDNN	✘	BIDMC: 99.16 %, MIMIC: 98.42 %, CapnoBase: 99.87 % Two-session MIMIC: 9 8.05 %, Biosec1: 89.26 %	EER: BIDMC (0.92), MIMIC (1.43), CapnoBase: (0.18) Two-session MIMIC (2.26), Biosec1 (11.42)
[10]	Butterworth filter, sliding window peak detection	SRNN	✘	–	Precision: Stress (85.82 %), WESAD (87.46 %); Recall: on Stress (85.41 %), WESAD (87.16 %); F-measure: Stress (85.16 %), WESAD (87.04 %)
[16]	Peak detection, distance-based thresholding, Z-score normalization	2 channel DNN	✔	–	FRR (1.37 %) and EER (0.68 %)
[33]	Butterworth filters applied to IR, green, and IMU channels with specific passbands	Deep sensor fusion Siamese network with triplet loss	✘	95 %	FRR at 0.01 FAR for 10-seconds: 0.12, FRR at 0.05 FAR for 10-seconds: 0.04
[8]	Fourth-order Butterworth filter, two-stage MA removal algorithm	RF, SVDD	✔	> 90 %	FAR (3.11 %), FRR (3.73 %), F1-score (97.2 %), recall (> 93.27 %),ADR(>93:33 %)
[66]	Median filter, Savitzky-Golay (SG) filter, regularization matrix, Smoothness prior approach	Ridge Regression	✔	μ 84 %	TRR (98 %), Variance (2.98 %)
[79]	low-pass filter, threshold that based on the energy accumulation	RF, SVM, DT, GB, K-NN	✘	> 90 %	F1-score (> 88 %), Pass Rate (96.67 % for legitimate users and 0.62 % for attackers at security level)
[74]	STE-Var, Peak-Valey	RF, SVM, KNN, NB, LR, MLP, GBDT	✘	94.9 %	–
[61]	Detrending, Peak Detection, Multicycle averaging	1D-CNN Siamese Network (3 convolutional layers + FC layers)	✘	97.23 % (N = 5 cycles)	AUC-ROC (0.990), EER (0.0357), FRR (≈ 0.2)
[77]	Butterworth band-pass filter, SMAF, VMD/CEEMDAN + FastICA	GBT	✔	90.73 % (static), 74 % (continuous MA after mitigation)	AFDR (3.9 %), ADR (88 %)

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Table 3 (*continued*)

R	Preprocessing Algorithms	Classifiers or algorithms	CV	Accuracy	Others indicators
[56]	Band-pass filter, optimal frame-size selection	Autoencoder neural network with multilayer perceptron structure	✔	97.9 %	EER (5.5 %)
[37]	Second-order Butterworth bandpass filter, moving average filter, Kullback-Leibler	SVM (RBF kernel)	✘	98.7 %	ADR (96 %), AFDR(8 %)
[48]	Savitzky-Golay filter, Skewness-DTW, Dynamic Programming, Binary Search	RF, SVM (RBF), GB, KNN, MLP, Feed-Forward NN, OnevsRest	✘	–	F1-Score (98 %)
[43]	Segment data	SVM(RBF)	✔	73 %	FAR (0.26 ± 0.12), FRR (0.26 ± 0.12)
[14]	–	Two cascading one-dimensional CNNs, with CNN-1 identifying users and CNN-2 verifying heartbeat consistency, optimized using the Adam algorithm	✔	99.27 %	Precision (73.20 %), recall (35.31 %), FAR (0.12 %), FRR (64.68 %)
[68]	Low-pass filter, polynomial fittings, linear transformation, peak detection	CNN	✘	BIDMC(99.5 %), MIMIC(99.6 %), CapnoBase(99.2 %)	FAR (BIDMC: 0.5, MIMIC: 2.4, CapnoBase: 0.8), FRR: (BIDMC: 0.4, MIMIC: 1.8, CapnoBase: 0.6)
[28]	Bandpass filtering, baseline removal, PCA, normalization, resampling, and artifact rejection	CVT-ConvMixer, a hybrid model combining Convolutional Vision Transformer (CVT) and ConvMixer with attention mechanism	✔	95 %	Sensitivity (97 %), Specificity (95 %), F1-score (95 %), AUC (0.96)
[73]	low-pass filter, beat separation algorithm, splits continuous PPG signals into individual beat	Homomorphic Random Forest (HRF)	✘	CapnoBase: 100 %, MIMIC: 92.5 %, BIDMC: 97.2 %, Oxford: 96.7 %, Self: 95.6 %	F1-score (CapnoBase: 100 %, MIMIC: 92.3 %, BIDMC: 97.2 %, Oxford: 96.4 %, Self: 95.3 %), EER (μ 2.14 %)
[2]	Bandpass filtering, noise filter	SVM, KNN, Naive Bayes, Ensemble Bagged Trees, GAM, CNN, LSTM	✔	99.8 %	EER (0.16), F1-Score (98.5 %), Sensitivity (97.1 %), Precision (97.1 %)
[13]	–	LSTM-based	✘	–	FPR (4.8285 %), FNR (4.8814 %)
[63]	Adaptive and notch filters	SVM, DT, RF, LR, MPL	✘	Pulse Transit Time PPG (100 %), BIDMC (99.7 %)	AUC (1.0), EER (0 %), Precision (1.0), recall (1.0), F1-Score (1.0)
[53]	Light strobe removal, heartbeat standardization scheme	SVM, DT, GB, LR	✘	94.52 %	F1-score (88.74 %), Precision (> 89.3 %), Pass Rate (legitimate user: 93.3 %, attacker: 3.11 %)
[51]	Direct current	Bi-LSTM, ADAM	✘	Bi-LSTM with raw signals, testing accuracy for 30 subjects (96.7 %), Bi-LSTM with feature-based data, testing accuracy (95.0 %). LSTM baseline with raw signals, testing accuracy (96.3 %). LSTM baseline with feature-based data, testing accuracy (59.1 %).	Precision (97.0 % raw signals, 30 subjects), Recall (100 % raw signals, 30 subjects), F1-Score (98.0 % raw signals, 30 subjects), FAR: (0.03 %), FRR: (0.00 R: (0.03 %)
[3]	Butterworth Bandpass filter, Amplitude normalization	LSTM	✘	98.79 %	Precision (98.19 %), Recall (98.14 %), and F1-measure (98.16 %)
[57]	Second-derivative PPG, Sliding, Sliding windows	Gaussian Mixture Models (GMM) with a diagonal covariance matrix	✘	Original subjects (89.1 %), Including the "Other" class (87.6 %)	std (11.7 %)
[50]	Fourth-order Butterworth bandpass filter, PSD, DC drift assessment	1D CNN and LSTM layers	✔	binary (99.64 %) Multisubject(98.8 %)	Precision (96.1 % binary, 98.56 % multisubject), Recall (94.63 % binary, 98.87 % multisubject) F1 Score (95.11 % binary, 98.59 % multisubject)

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Table 3
(*continued*)

R	Preprocessing Algorithms	Classifiers or algorithms	CV	Accuracy	Others indicators
[11]	Balancing, augmentation (flipping), Z-score normalization	BiLSTM with Adam optimizer	✘	95.26 %	–
[70]	PPG data augmentation	PulseID	✔	94.3 %	EER (2.9 %)
[12]	Savitzky-Golay filter	EfficientNetV2 B0 + LSTM	✘	99 %	Precision (99 %), Recall (99 %), F1-score (99 %)
[27]	4th-order Butterworth filter, LMS	CNN with SELU, dropout, ADAM optimization	✔	Single-session: IEEE TBME 2013 (100 %), TROIKA (91.7 %), Biosec1 (99.3 %), Biosec2 (98.7 %); Two-sessions: Biosec1 (86.1 %), Biosec2 (87.3 %)	train time (356s in IEEE TBME 2013), EER (IEEE TBME 2013: 0.1 %, Biosec2 Two-sessions: 12.6 %)

Note: CV: Cross-Validation. Abbreviations used: ROC (Receiver Operating Characteristics), ADR (Attack Detection Rate), AFDR (Attack False Detection Rate), EER (Equivalent Error Rate), FRR (False Rejection Rate), FAR (False Acceptance Rate), TRR (True Rejection Rate), FNR (False Negative Rate), TPR (True Positive Rate)

Finally, approaches in authentication are moving toward behavioral biometrics, where unique behavioral patterns are analyzed, such as handwriting dynamics through the analysis of dynamic changes in blood flow caused by hand movement during signing [48].

Advances in temporal modeling using recurrent and convolutional networks demonstrate significant technical progress but also reveal a tendency to prioritize complexity over practical justification. Although deep learning-based techniques offer innovative solutions for processing signals in diverse contexts, their marginal improvements in metrics (precision, F1, EER, etc.) compared to classical methods call into question their widespread viability. Furthermore, transforming 1D signals into 2D representations and employing hybrid architectures, while promising, introduces additional complexity that does not always yield substantial gains. Finally, the growing interest in behavioral biometrics highlights a path toward authentication based on unique patterns, emphasizing the need to balance technological innovation with efficiency and simplicity in future research.

4.3. Multimodality

The combination of multiple signals and/or sensors for authentication is a recurring strategy in the literature. The integration of various signals mitigates the inherent limitations of each particular modality.

The integration of PPG signals with additional physiological sources, particularly electrocardiogram (ECG) signals, has gained increasing importance for developing more resilient biometric systems by leveraging complementary cardiac patterns. In PPG-ECG fusion, various deep learning architectures have been proposed. Coelho et al. [14] present an approach based on cascading CNNs, where each signal is processed independently, and the resulting probabilistic representations are concatenated and combined by another CNN. However, this method does not perform true explicit multimodal fusion and is limited to sequential processing of the ECG and PPG signals. Similarly, Fan [16] proposes EPDF (A Dual-Channel Individual Classification and Fusion of ECG and PPG Signals in Wearable Devices), a dual-channel design that employs MultiRes blocks and Spatial Pyramid Pooling (SPP) to extract multiscale features from both signals. Although this approach incorporates fusion with self-attention mechanisms and dual concatenation, preserving information, it introduces unnecessary architectural redundancy. Both approaches lack a solid methodological justification for their integration strategies, particularly in handling temporal asynchronies and scale differences among the biomedical signals.

Along similar lines but outside deep learning strategies, Ahamed et al. [2] investigate an initial fusion of signals (RMS, geometric mean) and derive features in both the time and frequency domains (Shannon entropy, AR coefficients) to address vulnerabilities such as device spoofing. However, the fusion yielded minimal improvement compared to using ECG signals alone, calling into question the effectiveness of the fusion method. One unpredictable characteristic missing in the previously reviewed works on cancelability is addressed by Siam et al. [63], who present a cancelable ECG and PPG fusion method using the discrete cosine transform (DCT) and the Elazm Table (E-Table). Nevertheless, this method does not provide a mathematical justification or an analysis of its cryptographic properties to determine the effectiveness of the cancelability.

Beyond combining PPG with ECG, the fusion of PPG with other sensor types is being explored to improve authentication. Kumar et al. [33] present a system for wrist-worn devices that integrates PPG signals and motion sensor data (IMU) using a deep Siamese network, highlighting its ability to incorporate new users with only a few examples. The reliance on motion data alongside PPG signals could introduce potential vulnerabilities, as movement patterns may be easier to replicate or manipulate compared to purely physiological signals. Similarly, Pan et al. [53] integrate PPG signals with vibrations produced by an embedded motor, incorporating high-frequency components as an additional biometric signature. The system employs noise reduction measures, random challenges to thwart replay attacks, and ongoing model updates to ensure sustained performance over time. However, while the incorporation of motor vibrations provides a novel layer of security, reliance on hardware-specific components such as the motor may limit adaptability across different devices, particularly in low-end wearables where such components may not be readily available.

These works collectively underscore the trend of multimodal fusion. The field of multimodal biometric authentication is rapidly evolving thanks to the creative integration of physiological signals such as PPG and ECG, as well as other complementary sources such as motion sensors or induced vibrations. Although advances in deep learning architectures and fusion methods have shown great potential for improving system accuracy and robustness, significant challenges remain that must be addressed. Among these are managing temporal asynchronies between signals, adapting to devices with limited resources, and providing more robust methodological justifications for the proposed fusion strategies. Furthermore, ensuring the security and privacy of biometric data remains a critical priority, particularly in applications where cancelability is essential.

4.4. Multifactor authentication (MFA)

Multi-factor authentication (MFA) enhances security by incorporating two or more authentication factors (such as a second signal, biometrics, or a credential). This method effectively addresses the limitations of systems dependent on a single authentication factor and

substantially improves security, as an attacker must circumvent all factors.

Studies use PPG signals as a primary or complementary authentication factor alongside another factor, providing an additional layer of biometric security [8,66]. The multi-factor approach significantly enhances the robustness of authentication systems, particularly when integrated with cancelable feature template techniques [8], which enable the secure revocation and renewal of biometric credentials in the event of compromise.

Behavioral biometrics based on the PPG signal, obtained by analyzing dynamic patterns in the interaction between the user and the system, represent a more robust authentication method. Recent research indicates that PPG patterns not only convey physiological information but also integrate distinctive behavioral traits, such as pressure patterns [74] or characteristic gestures [79]. This expands the set of identifiable attributes that can be utilized in multi-factor authentication systems.

It is essential to note that the combination of PPG with behavioral biometrics in the context of multi-factor authentication (MFA) holds considerable potential. However, further research is needed to standardize these gestures and ensure their consistency across different sessions. Similarly, it is necessary to verify the reliability of each factor individually, and in the case of the PPG factor, to ensure that it is both cancelable and robust.

4.5. Continuous authentication

Continuous authentication verifies a user's identity throughout an entire session, which is essential for high-security applications. As dynamic, unobtrusive signals acquired in real time via wearable devices, PPG is ideally suited for this purpose. This evolving field employs diverse techniques to achieve reliable authentication.

One of the main challenges in continuous authentication is signal quality and the presence of motion artifacts (MAs), as these interfere with the primary objective: maintaining session continuity. Motion artifacts can disrupt a legitimate user's session. Various efforts have focused on mitigating these effects. For example, Li et al. [37] use intra-aural signals, which are prone to artifacts, and propose a wearable pipeline that includes preprocessing and motion artifact removal using Kullback-Leibler (KL) divergence. Nevertheless, accuracy is still affected by motion artifacts, dropping from 98 % to 76 %. Another approach is presented by Zhao et al. [77], who introduce TrueHeart, which detects corrupted segments using statistical metrics (kurtosis, skewness, standard deviation) and applies techniques such as CEEMDAN (Complete Ensemble EMD with adaptive noise) and FastICA to mitigate artifacts. They also acknowledge that other factors affect signal quality, such as sensor characteristics, light interaction with melanin (ethnicity), emotional state, and energy consumption, adding further complexity to continuous authentication. In this context, Lien et al. [43] introduce eigenhearts, a feature derived via PCA that captures unique temporal patterns of heart rate (HR), with special emphasis on robustness against demographic variables such as sex, age, and BMI (Body Mass Index). While this work represents a significant advance in addressing diversity, its real-world applicability is limited, achieving accuracies below 80 % and offering limited demographic representation.

On the other hand, privacy is a crucial aspect of continuous authentication. Zhang et al. [73] combine homomorphic encryption (CKKS) with an encrypted Random Forest model (HRF), enabling remote authentication without compromising data confidentiality. The approach includes rigorous feature selection, reducing 541 features to 19 via PCA and mRMR, thereby ensuring both efficiency and security.

There is a trend toward increasingly complex deep learning architectures, ranging from autoencoders that transform PPG signals into a discriminative latent space [56], to multi-scale (time, frequency, wavelet) fusions with CNNs [68], and to the use of LSTMs that predict future PPG signal values [13] for driving applications. While these approaches aim to improve model accuracy, it is important to evaluate their effectiveness across various scenarios and determine which methods are best suited for each implementation.

Together, these works reflect a diverse ecosystem of methods centered on PPG as the core of continuous biometric authentication. The proposals range from sophisticated preprocessing strategies and deep learning-based models to cryptographic security mechanisms and multimodal fusion, all aimed at optimizing accuracy, privacy, computational efficiency, and applicability in real-world contexts such as wearables, smartphones, and vehicles. However, challenges such as motion artifacts, ethnicity, emotional state, and health status must be addressed as priorities to ensure a robust authentication system.

4.6. Sensors and hardware platforms for data collection

One prominent aspect evident in the surveyed literature is the prevalent reliance on proprietary datasets, which underscores the wide range of hardware options used for signal acquisition. As outlined in Table 4, platforms such as the Arduino UNO and STM32 frequently serve as the foundation for developing affordable, tailored prototypes, commonly paired with specialized PPG sensors from the MAXIM Integrated lineup (such as the MAX30101, MAX30102, or MAX30105). Certain investigations also employ off-the-shelf devices, including the Fitbit Sense, Empatica E4, or smartphone cameras such as those in the iPhone X, leveraging their embedded sensors to gather data in everyday, less structured settings. Furthermore, it is common to incorporate supplementary sensors to enhance datasets; for instance, Inertial Measurement Units (IMUs) equipped with accelerometers and gyroscopes such as the LIS2DH12 or MPU-6050 are often added to reduce motion artifacts or to record behavioral biometrics. Some creative strategies even involve vibration motors to introduce additional stimuli that augment the biometric signals. This assortment of hardware setups reflects the varied objectives pursued across these studies.

Although this diversity in hardware highlights the adaptability of PPG technology, it also creates a lack of standardization that, together with the proprietary nature of most datasets, poses substantial barriers to progress in the field. Variations in sensor attributes such as sensitivity, sampling rates, and noise characteristics bind the outcomes of any given study closely to its specific acquisition tool. This limitation significantly undermines the reproducibility and broader applicability of developed models, as an algorithm refined on data from a bespoke prototype

Table 4
Platforms and sensors used by studies for signal acquisition.

R	Platform	PPG Sensor	Other Sensors
[50]	nRF52832	AFE4900	
[8]	Intel Core i7 CPU	MAX30105	
[66]	STM32	MAX30101	LIS2DH12
[79]	Arduino UNO		MPU-6050
[74]	Arduino UNO	MDL0025	
[77]	Arduino UNO		
[37]	Arduino		
[48]	Arduino UNO	PPG sensor by World Famous Electronics	
[43]	Fitbit Sense	Fitbit Sense Sensor	
[73]	PC	Camera	
[13]	Empatica E4	Empatica E4 Sensor	
[12]	Arduino UNO	MAX30102	
[53]	Arduino UNO	NI USB-6002	3610# vibration motor module
[51]	iPhone X	Camera	
[57]	Arduino ATmega328P		

might underperform when applied to signals from an alternative commercial device, unless such a cross-comparison is explicitly conducted. The problem is compounded because the acquired data are seldom shared publicly, obstructing equitable evaluations among diverse authentication frameworks and compelling each research group to allocate resources toward independent data collection. Consequently, the field remains fragmented, with performance indicators that are difficult to interpret meaningfully, ultimately impeding the evolution of these systems from experimental lab models to reliable, practical applications.

4.7. Key techniques: preprocessing, feature extraction, and classification

As shown in Table 3, the feature extraction methods vary widely, with convolutional neural networks (CNN) being the most common approach, used in [14,27,33,50,61,70,71,75] to derive features from PPG signals. Time domain (TD) features are also prevalent, extracted in [2,45,48,68,79], reflecting their usefulness for capturing temporal patterns. Frequency domain (FD) features appear in [2,10,45,51,68,79] often complementing TD analysis. Other methods, such as fiducial points (FP) [8,73], specialized techniques such as miniROCKET-based extraction [66], multiwavelet decomposition [56], MultiRes Blocks [16], 2D scalograms [28], MFCCs [63], or LSTM-based methods [3,11] are described in individual studies, highlighting the diversity of approaches tailored to specific classification tasks.

Motion artifact (MA) mitigation is a common practice, applied in [3,8,16,27,28,33,37,45,48,56,66,73,77,79], underscoring its importance for improving signal quality.

In the classification domain, supervised algorithms predominate, with SVM and CNN standing out with an equal number of implementations. On one hand, SVMs were employed in [37,43,48,53,57,61,63,73], while on the other hand, CNNs were used in [8,14,27,28,57,68,75]. However, cross-validation has only been applied in [2,8,14,16,27,28,43,50,56,66,70,77], questioning the robustness and generalizability of the results, despite reporting high accuracy (above 90 % in [3,11,12,33,37,45,48,50,51,53,61,68,71,74,75,79], with maximums of 100 % accuracy [63,73]). In addition to classic metrics such as precision, recall, and F1-score, specific security indicators (FRR, FAR, EER) are frequently reported. However, their variability and the general lack of testing under adversarial conditions reveal that a high accuracy in classification does not necessarily imply a robust security posture. This critical gap in the literature, where security is often overlooked or evaluated superficially, will be the central focus of the following section.

5. Main attacks and their relevance in PPG-based authentication systems

Analysis of the various components of PPG-based authentication systems reveals a gap in the literature regarding to security (Table 2). Only [2,8,53,66,74,77,79] provide a comprehensive evaluation of attacks, highlighting Random Attacks [2,8,53,79], Replay Attacks [8,53,74,79], Emulating Attacks [66], Presentation Attacks [74], Synthesis Attacks [53,77,79], Spoofing Attacks [2], Tampering Attacks [2], and Registration Phase Attacks [2]. The remaining studies only briefly mention security or omit attack analysis entirely. This situation may arise because current research on PPG-based authentication has predominantly focused on classification (binary or multiclass). However, this approach often overlooks critical security considerations that are essential for practical implementation.

A robust biometric system should not only be accurate in discriminating between legitimate users, but also incorporate cancelability mechanisms, a strategy implemented only by [8,63,73], as well as proactive measures that mitigate deliberate threats, such as identity spoofing, injection of falsified signals, or manipulation of physiological data. Cancelability is central to safeguarding the long-term usability of biometric systems because, unlike passwords or tokens, biometric traits cannot be changed if compromised. Therefore, cancelable biometrics enable the transformation of the original template into a revocable form, ensuring that, if the stored data is exposed, a new transformed version can be generated without compromising the uniqueness of the individual. This property provides resilience against template theft, prevents cross-matching across different systems, and aligns with privacy-by-design principles, which are increasingly required in modern biometric applications. Despite its significance, cancelability remains one of the least explored areas in PPG-based authentication, highlighting a critical gap in the current body of knowledge.

These vulnerabilities are particularly relevant in PPG-based systems, since the signals, being acquired through noninvasive sensors, present a higher risk of being intercepted, replicated, or altered through replay attacks or through the generation of synthetic patterns.

In addition to the attacks reported in Table 2, this section presents the results of an extensive search aimed at identifying the main attacks that directly affect PPG-based authentication, based on a review of specialized cybersecurity literature. It is important to note that the attacks described correspond exclusively to those addressed in articles with a primary focus on security analysis.

5.1. Attack classification

Establishing a comprehensive security analysis framework is essential to identify, understand, and mitigate attacks targeting PPG-based biometric systems. The threat landscape of these systems is complex and multifaceted, necessitating a thorough examination of both attack vectors and their potential consequences. According to the seminal work of [4], threats to biometric systems can be broadly classified into two main categories: passive attacks and active attacks.

1) Passive attacks refer to threats in which adversaries do not directly manipulate the biometric system but instead intercept, monitor, or eavesdrop on data as it travels through the system. The primary concern in this category is the interception of sensitive biometric information during transmission, particularly when signals are sent through unencrypted channels. The absence of cryptographic protection renders PPG data highly vulnerable to unauthorized access and observation. Such passive attacks not only compromise user privacy but may also enable subsequent active attacks by providing adversaries with valid biometric data for analysis and replication. These risks are particularly pronounced in wireless transmission environments, where PPG signals may be captured by malicious actors within the broadcast range.

2) Active attacks, by contrast, involve direct interference with the biometric authentication process, in which adversaries attempt to manipulate, inject, or bypass the system to gain unauthorized access. These attacks pose a more immediate and severe threat, as they directly compromise the integrity of the authentication mechanism. In the context of PPG systems, active attacks can be further categorized into several subtypes, each necessitating tailored countermeasures:

- Presentation attacks: The use of physical or digital replicas of biometric signals.
- Spoofing: The emulation of PPG signals through synthetic or fabricated devices.
- Replay attacks: The reuse of previously captured legitimate signals to deceive the system.

Security systems should be evaluated not only for performance but also for security, as it is a fundamental characteristic of biometric authentication systems. Unlike passwords or PINs, which can be reset after compromise, a stolen PPG biometric signature cannot be replaced. This creates long-term vulnerabilities for user security and privacy, as a single breach may permanently undermine the effectiveness of this method. Consequently, highly robust and proactive detection and mitigation mechanisms are required to safeguard biometric data. These challenges are further amplified as the field advances toward continuous authentication approaches, where persistent and adaptive active attack strategies must be anticipated and addressed.

5.2. Presentation attacks

Presentation attacks (PA) pose a significant risk to PPG-based authentication systems, primarily due to the widespread availability of sensors in wearable devices. Recent research classifies these attacks based on intent, distinguishing between non-subversive cases, such as non-malicious errors caused by motion artifacts [26], and subversive cases, which involve deliberate spoofing using synthetic signals or compromised sensors [26].

According to [22], PPG-based systems are particularly susceptible to these subversive attacks, as an attacker can capture a victim's PPG signals via an external sensor, relay fraudulent data to the authentication system, and exploit the absence of sensor location verification, even using signals from unauthorized body parts [23,24]. In experimental evaluations, Hinatsu et al. [22] demonstrated that such attacks could achieve a success rate exceeding 80 % under ideal conditions. Specifically, when sensors were stabilized across all measurement sites, success rates of up to 1.000 and 0.950 were obtained for different waveform estimation methods. These results underscore the high probability of compromise in PPG-based authentication systems when adequate countermeasures are not implemented.

This vulnerability is visually represented in Fig. 3, which illustrates a presentation attack scenario in which an attacker compromises or impersonates an authorized wearable device (depicted as a wristband emitting a heartbeat signal). The attack results in the injection of an "unreal signal," that is, a fabricated or manipulated PPG waveform directly into the authentication system, thereby bypassing legitimate verification and potentially granting unauthorized access.

To reduce these risks, recent studies have proposed defenses such as waveform analysis techniques that verify signal origin through distinctive amplitude and frequency characteristics [20], and liveness detection methods employing the Hilbert transform and weighted K-nearest neighbors (K-NN) algorithms to assess the physiological authenticity of signals [17]. Nevertheless, many existing systems still lack built-in safeguards against signals from unverified locations, leaving this vector as a persistent vulnerability [24].

5.3. Spoofing attacks

PPG spoofing attacks involve the systematic generation of synthetic signals that replicate the characteristics of legitimate users [18]. By gaining remote access to authentic signals, attackers can generate fake signals to deceive authentication systems. As illustrated in Fig. 4, these spoofing attacks follow a two-stage process: (1) false signal generators, such as Generative Adversarial Networks (GANs), create multiple counterfeit PPG waveforms with varying characteristics, and (2) these fabricated signals are subsequently presented to authentication systems to bypass security measures.

A particularly concerning development is remote PPG (rPPG), which derives physiological signals from video recordings [21,65]. This technique enables spoofing without direct physical access to the sensor [21,39,60,72], significantly expanding the attack surface. While [38] introduced video-based defenses, their method remains vulnerable to unauthorized recordings.

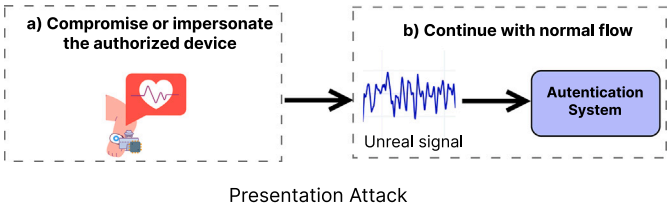


Fig. 3. PPG presentation attack workflow. a) An attacker compromises or impersonates the authorized wearable device (e.g., a wristband sensor). b) Inject an unreal or fabricated signal into the authentication system.

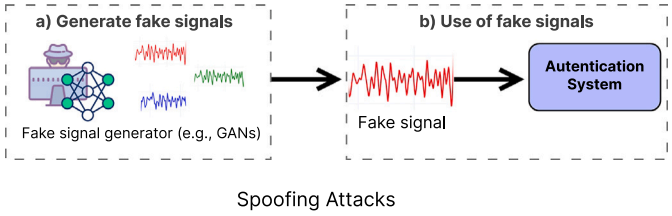


Fig. 4. PPG spoofing attack workflow: a) A false signal generator (e.g., GAN) produces synthetic PPG waveforms. b) The generated fake signals are used to compromise biometric authentication systems.

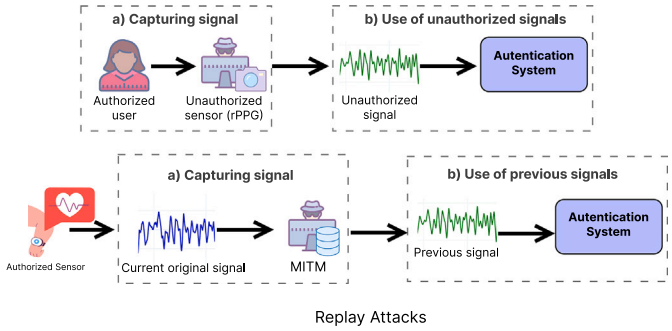


Fig. 5. Overview of replay attack vectors in PPG authentication. Top: Unauthorized sensor (e.g. rPPG) captures signals from authorized users for subsequent replay. Bottom: MITM attack intercepts current signals from authorized sensors and replays previously stored signals to gain unauthorized access.

Furthermore, most existing research focuses on facial spoofing [19,30,31], overlooking the distinctive challenges associated with PPG signal authentication. Multimodal approaches have emerged as a promising countermeasure. For instance, Poretti et al. [55] integrated PPG with electrocardiogram (ECG) signals through neural networks, substantially reducing the effectiveness of unimodal attacks. However, this approach increases system complexity and computational requirements, and its feasibility for deployment in commercial devices remains unexamined.

5.4. Replay attacks

Replay attacks exploit the reuse of authentic PPG signals captured from previous authentication sessions. As depicted in Fig. 5, these attacks manifest in two primary forms: (1) rPPG-based replay attacks, in which an adversary uses an unauthorized sensor (such as a remote camera) to capture the PPG signal of an authorized user and subsequently replays this unauthorized signal to the authentication system [38], and (2) Man-in-the-Middle (MITM) attacks, in which an attacker intercepts legitimate signals transmitted from an authorized sensor, stores them in a database, and later retransmits these previous signals to gain unauthorized access. Unlike spoofing attacks, replay methods do not require the generation of synthetic signals; instead, they compromise the integrity of the communication channel between the sensor and the authentication system.

Effective mitigation strategies for replay attacks require a multi-layered defense approach. Communication encryption serves as the foundational layer, employing robust cryptographic protocols (e.g., TLS 1.3, AES-256) to preserve both the integrity and confidentiality of transmitted physiological data. Complementing this, temporal validation mechanisms including timestamps, nonces, and session-specific cryptographic challenges prevent the reuse of previously captured signals by ensuring that each authentication attempt is temporally bound and unique. Anomaly detection algorithms constitute another critical component, employing machine learning techniques to identify atypical patterns in signal transmission characteristics, such as unexpected temporal

gaps, statistical inconsistencies, or transmission anomalies indicative of replay attempts. Finally, session verification mechanisms guarantee the uniqueness of each authentication process through cryptographic session tokens and signal freshness validation, explicitly rejecting any signal segments that have been utilized in prior sessions.

Despite the prevalence of these attack vectors, the literature reveals a significant gap in comprehensive evaluation frameworks that systematically address replay attacks in PPG-based authentication systems. This deficiency highlights the critical need for a standardized assessment methodology.

6. Evaluation of PPG-based authentication systems

While the taxonomy proposed in the previous section is useful for an initial classification and facilitates the analysis of different PPG-based authentication approaches, it is necessary to consider the coexistence of multiple characteristics within the same system (for example, a system can simultaneously implement continuous and multimodal authentication).

To address this complexity and provide a structured view of the current landscape regarding the intersections between approaches including behavioral biometrics due to its relevance Fig. 6 presents a Venn diagram that classifies different PPG-based authentication systems according to four fundamental characteristics: continuous (red circle), multifactor (yellow circle), behavior-based (blue circle), and multimodal (green circle). The intersections between the circles explicitly illustrate the overlap and combination of these characteristics in the various authentication approaches identified in the literature.

Based on these interactions, this section presents recommendations for evaluating authentication systems. The objective is to standardize evaluation indicators, propose recommended datasets, and outline security considerations that should be addressed when assessing systems with these characteristics.

6.1. Datasets

Our thorough review revealed a notable characteristic: many studies rely on custom datasets, employing their own signal acquisition devices [8,12,13,33,37,43,48,50,51,53,57,66,73,74,77,79]. While this practice is justified in emerging areas such as behavioral biometrics, where standardized datasets are scarce, it highlights a critical challenge for the field: the lack of public, standardized benchmarks hinders direct comparison and reproducibility. This reliance on proprietary data collection also raises important ethical and privacy concerns regarding the sharing of physiological biometric data, a point that will be elaborated on later. Therefore, transparency from researchers regarding their acquisition tools, sensors, platforms, participant conditions, and activities is paramount, as is the public disclosure of data whenever feasible.

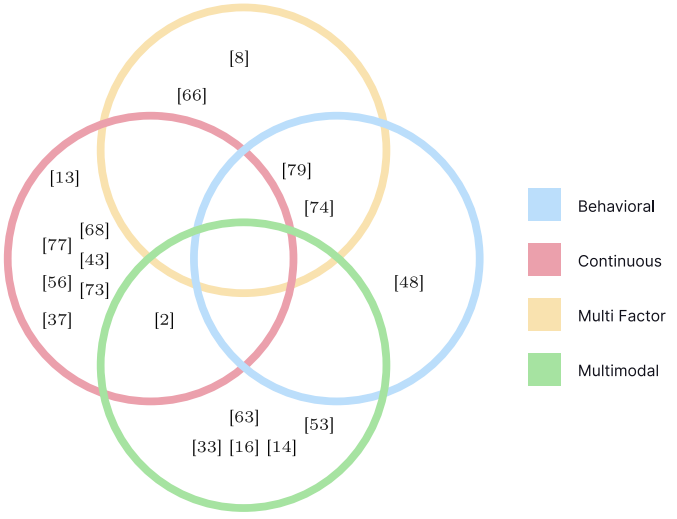


Fig. 6. Overview and relationships among PPG-based biometric authentication approaches.

The primary signal acquisition modality is via wrist-worn devices [8,13,33,43,48,53,66,74,77,79], followed by finger-based sensors [12,50,51,57], with a minority using face [73] or ear [37] locations. This diversity in acquisition hardware, particularly between consumer-grade wearables (e.g., wristbands) and medical-grade sensors (e.g., fingertip devices), introduces a crucial challenge for multi-platform generalization. Signals can vary significantly in quality and characteristics depending on the device, posing a substantial obstacle to developing robust, cross-device authentication models.

Furthermore, an explicit comparison of dataset demographics (age, ethnicity, health conditions) is essential to understand and mitigate biometric bias. Many existing datasets lack the diversity required to build equitable systems. As shown in Fig. 7, which details the frequency of dataset usage, 17 of the 32 analyzed works employ proprietary datasets, while the remainder use medical datasets. Without standardized demographic reporting, it is difficult to assess whether a model's performance degrades for specific demographic groups.

In response to these challenges, recent years have seen the development of datasets specifically designed for authentication. Table 5 provides a comparative overview to inform dataset selection, detailing the number of participants, access conditions, capture duration, environment, participant diversity (gender, age, ethnicity), and the capture device. For instance, Aathreya et al. [1] introduced a multimodal dataset for continuous authentication under induced emotional contexts. Lien et al. [43] proposed a dataset collected longitudinally via commercial wearables in natural environments, representing a step toward capturing real-world variability. Addressing signal quality directly, Ho et al. [25] presented WF-PPG to study the impact of contact pressure on wrist PPG signals. To tackle long-term variability, Li et al. [40] proposed SigD, emphasizing cross-session scenarios. Finally, Meier et al. [47] collected WildPPG in uncontrolled conditions, providing valuable data for assessing generalization across diverse real-life situations.

In summary, the field must move beyond isolated data collection efforts. Future work should prioritize the creation and adoption of public datasets that are not only large but also demographically diverse and acquired under varied, realistic conditions. This represents the most effective path toward reducing biometric bias and improving the generalization and ethical standing of physiological biometric authentication systems.

6.2. Performance metrics

A critical challenge in the reviewed studies is that PPG-based authentication is predominantly treated as a classification problem, leading to an overreliance on standard classification metrics (see Table 3). However, biometric authentication systems must be validated across multiple dimensions, particularly reliability, usability, and operational efficiency, rather than solely on classification accuracy.

Our review (Fig. 8) reveals that the most frequently reported metrics include Accuracy, F1-score, EER, Precision, and Recall, whereas biometric-critical authentication metrics such as FAR, FRR, and ROC curve appear less consistently. Metrics related to usability (e.g., response time, enrollment success rate) and efficiency (e.g., energy consumption, training time) are rarely reported. More critically, there is a notable lack of standardization in how these metrics are measured and reported, especially in realistic deployment scenarios.

To enable meaningful comparisons and ensure robust system evaluation, we recommend the following prioritization framework:

- Priority 1: Basic biometric authentication metrics [64]

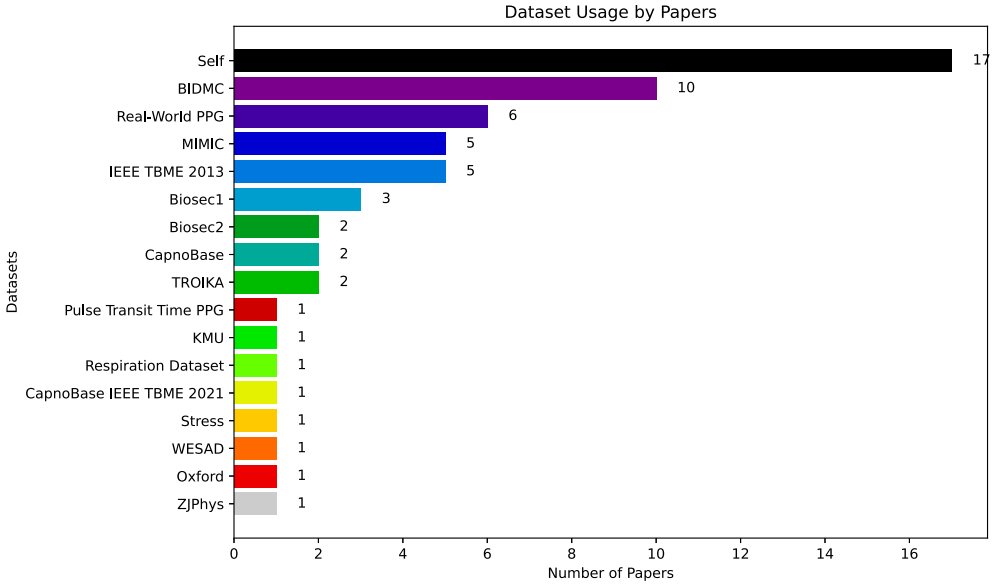


Fig. 7. Frequency of use of different datasets employed by various studies.

Table 5
Proposed datasets for future research.

R	User's Number	Accessibility	Time	Collection environment	Data diversity	Device
[43]	27	Data will be made available on request	Four to Five days	Natural environments capturing various daily life activitie	15 females, 12 males, ages 18–63, collected basic demographics, including sex, age, and Body Mass Index (BMI), from each user	Fitbit Sense
[1]	27	Data will be made available on request	Three sessions with an interval ranging from a few days to weeks between each session	Controled, with emotional stimulation	10 females, 17 males, eges 8–72, Ethnic groups (Asian, White, Middle Eastern/North African, African American, Hispanic/Latino)	Empatica E4
[25]	27	WF-PPG	Six sessions; four-minute data recording	Controled	11 females, 16 males, ages 20–32, controlled variation of the contact pressure on the wrist sensor	MAXM86161
[40]	232	SigD	More than two data collection sessions at different times. ≈ 5.7 min.	Clinical	49 % females, 51 % males, ages 21–89, intervals between sessions of 3 hours to 3000 days	
[47]	16	WildPPG	average duration of 13.5 hours	Round trip from Zurich to Jungfraujoch, with multiple activities	Multiple body locations	MAX86141, SFH7072(ams-OSRAM)

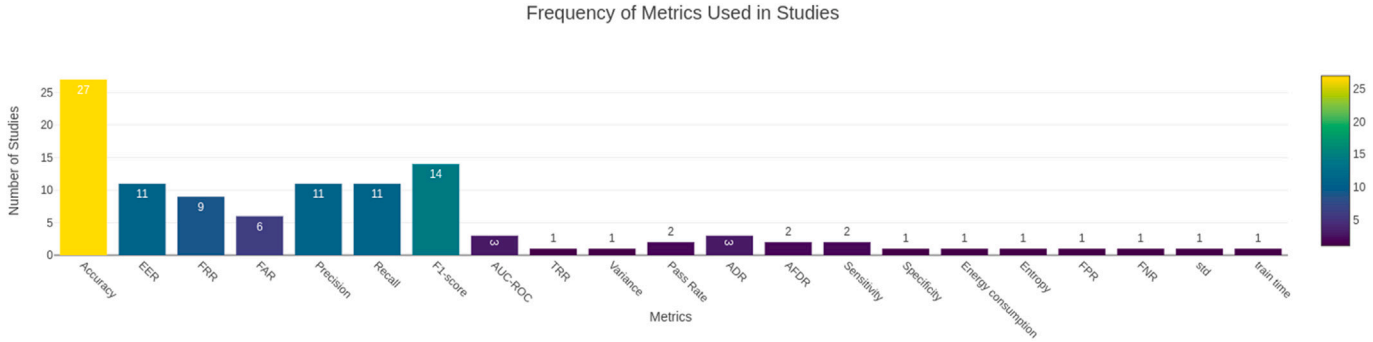


Fig. 8. Frequency of metrics used in PPG-bases authentication and identification systems.

1. False Acceptance Rate (FAR): Measures how often unauthorized users are incorrectly authenticated.
2. False Rejection Rate (FRR): Measures how often legitimate users are denied access, which impacts usability.
3. Equal Error Rate (EER): The operating point where FAR equals FRR, providing a single-value security performance indicator.
4. Genuine Acceptance Rate (GAR): GAR is the genuine acceptance rate, calculated as the complement of the FRR ($1 - FRR$).
5. ROC Curve: Visual representation of the trade-off between GAR (Genuine Acceptance Rate) and FAR across different decision thresholds.

- Priority 2: Classification performance (cross-validation results preferred)

1. Accuracy: Overall correct classification rate for general validation.
2. F1-score: Harmonic mean of precision and recall, especially important for imbalanced datasets.
3. Precision and Recall: For detailed analysis of system behavior.

- Priority 3: Operational viability

1. Latency: Average time taken to process an authentication request.
2. Throughput: Number of authentication attempts processed per unit time.
3. Failure to Enroll Rate (FER): Percentage of users unable to register successfully.

- Priority 4: Resource efficiency (important for wearable/mobile implementations)

1. Energy consumption per authentication.
2. Model size and memory footprint.
3. Training time and computational requirements.

The evaluation of PPG-based biometric authentication systems is currently affected by an overemphasis on standard classification metrics. For the field to advance toward practical and secure implementations, it is imperative to adopt a comprehensive evaluation framework that prioritizes biometric authentication metrics such as FAR, FRR, and EER, complemented by assessments of usability and operational efficiency. Standardizing the measurement and reporting of these metrics, particularly in realistic deployment scenarios, constitutes a critical step to enable meaningful comparisons and ensure the development of robust and reliable biometric systems.

6.3. Security evaluation criteria

Security systems should be evaluated not only in terms of performance, but also in terms of security, as security constitutes a fundamental characteristic of biometric authentication systems. Therefore, providing rigorous security evaluations in studies should be considered standard practice. This article already details the most common attacks reported in the literature to which PPG-based authentication systems are susceptible. To ensure that future research incorporates a robust security posture, evaluations should include several fundamental components: an explicit security analysis to identify potential vulnerabilities, a clear description of the implemented countermeasures to mitigate these threats, a comprehensive evaluation under attack scenarios to test system resilience, and detailed information on the mechanisms for protecting biometric templates.

In this regard, few works mention or analyze how their approach resists certain attacks; some rely on assumptions in their methods but provide no evidence to support resilience against these attacks [53,74]. Others, while presenting a “Threat Modeling” exercise, often do so in a cursory manner and based on assumptions [79]. Some works offer a more thorough security perspective [66], but still lack all elements necessary for an accurate evaluation. In this context, the adversary model proposed by Cao et al. [8], not only considers attacks but also relies on metrics and strategies to protect biometric templates, such as cancelability. Overall, only 4 of 32 reviewed papers explicitly present security as a central focus of their research; the remainder rely on even vaguer assumptions regarding security.

Given the identified issues in the security perspective, the following security assessment framework is proposed:

6.3.1. Vulnerability assessment

We begin a thorough security analysis with threat modeling to systematically identify potential vulnerabilities in PPG-based systems. Common models such as STRIDE (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege) can be applied to map risks across the system lifecycle, from signal acquisition to template matching. For PPG systems, key vulnerabilities include:

- **Signal interception**: During transmission from wearable devices to authentication servers, unencrypted PPG signals can be eavesdropped, especially in wireless environments.
- **Template exposure**: Stored biometric templates, if not properly protected, are susceptible to theft, leading to irreversible compromise since PPG traits cannot be changed such as passwords.
- **Adversarial machine learning**: Deep learning models used for feature extraction or classification can be fooled by adversarial examples, such as perturbed PPG waveforms designed to mimic legitimate users.
- **Hardware dependencies**: Reliance on specific sensors (e.g., wrist-worn devices) introduces risks from device tampering or substitution with malicious hardware.

Vulnerabilities should be prioritized based on their likelihood and impact, using quantitative methods such as Common Vulnerability Scoring System (CVSS) scores.

6.3.2. Evaluation under attack scenarios

Security assessments should simulate realistic attack scenarios to measure system resilience. Some possible attack scenarios include the following:

- **Presentation attack testing**: Use synthetic PPG generators (e.g., GANs) to create fake signals and assess detection rates, following ISO/IEC 30,107 guidelines [29]. Report metrics such as Attack Presentation Classification Error Rate (APCER) and Bona Fide Presentation Classification Error Rate (BPCER) [49].
- **Replay and spoofing simulations**: Replay captured signals via man-in-the-middle setups or spoof using rPPG from video recordings, evaluating success rates under varying conditions (e.g., different lighting or motion levels).
- **Adversarial robustness testing**: Apply perturbations to input signals using techniques such as Fast Gradient Sign Method (FGSM) and measure model degradation.
- **Cross-dataset and cross-device evaluation**: Test generalization by attacking models trained on one dataset/device using attacks from another, highlighting hardware-specific vulnerabilities.

Evaluations should include both white-box (attacker knows system details) and black-box scenarios, with results stratified by attack type and intensity.

6.3.3. Countermeasures and mitigation strategies

For each identified vulnerability, explicit countermeasures must be described and justified. Effective strategies for PPG-based systems include:

- **Encryption and Secure Communication**: Implement end-to-end encryption (e.g., TLS 1.3) for signal transmission and homomorphic encryption for processing encrypted templates without decryption, as explored in [73].
- **Liveness Detection**: Integrate mechanisms to verify signal authenticity, such as analyzing waveform variability, inter-pulse intervals, or combining PPG with accelerometer data to detect motion inconsistencies indicative of replays or spoofs.
- **Cancelable Biometrics**: Transform PPG templates into revocable forms using techniques such as discrete cosine transform (DCT) or biohashing [63], allowing template renewal in case of compromise without retraining the entire system.
- **Multimodal Fusion**: Combine PPG with complementary biometrics (e.g., ECG or IMU data) to increase attack complexity, while ensuring fusion does not introduce new single points of failure.
- **Anomaly Detection**: Employ machine learning-based monitors to flag unusual patterns, such as sudden changes in signal quality or transmission latency that may indicate tampering.

Countermeasures should be evaluated for their overhead on system performance, ensuring they maintain usability in resource-constrained wearables.

6.3.4. Biometric template protection

A core component of security analysis is detailing template protection mechanisms. PPG templates must satisfy key properties:

- **Diversity**: Templates should not be reusable across applications to prevent cross-matching.
- **Cancellability**: Ability to issue new templates if compromised.
- **Unlinkability**: Ensure transformed templates cannot be linked back to the original user.
- **Performance Preservation**: Protection should not significantly degrade authentication accuracy.

6.3.5. Security-specific metrics

Based on the reference work of [76], the following security metrics are proposed:

1. Attack Detection Rate (ADR)

The ADR in PPG-based biometric authentication measures the system's effectiveness in correctly identifying intrusion attempts or impostors (using false signals). It is calculated as the number of correctly detected attacks (N_{ca}) divided by the total number of attacks (N_{ta}), evaluating robustness against fraud in environments such as wearables.

$$\mathrm{ADR} = \frac{N_{ca}}{N_{ta}} \tag{3}$$

2. Attack False Detection Rate (AFDR)

The AFDR in PPG-based biometric authentication measures the proportion of legitimate users incorrectly classified as attackers. It is calculated as the number of false attackers (N_{fa}) divided by the total number of users (N_{tu}), evaluating the system's accuracy in minimizing unfair rejections in wearables or cardiac monitoring.

$$\mathrm{AFDR} = \frac{N_{fa}}{N_{tu}} \tag{4}$$

3. Receiver Operating Characteristic (ROC) Curve

The ROC curve visualizes the trade-off between the ADR and AFDR. The optimal operating point is the closest to the top-left corner, minimizing both false negatives and false positives. The distance D from any point on the curve to this ideal corner is given by:

$$D = \sqrt{(1 - \mathrm{ADR})^2 + (\mathrm{AFDR})^2} \tag{5}$$

In this systematic review, only two studies incorporated the ADR metric [8,37], highlighting an opportunity to standardize evaluation in PPG-based biometric authentication. Future research is encouraged to adopt not only the ADR but also complementary metrics such as AFDR and ROC, aligned with the standard definitions of the various authentication systems described in the taxonomy. This adoption would facilitate more robust comparisons between studies, improve the evaluation of effectiveness against imposters in environments such as wearables and IoT, and promote advances in the robustness and accuracy of systems.

7. Discussion

This systematic review highlights the rapid advancement of PPG-based authentication, revealing a domain characterized by significant technological progress yet hindered by fundamental limitations. Our examination reveals a clear shift toward sophisticated deep learning architectures and multimodal integrations, all aimed at enhancing classification precision. Nonetheless, the intensive emphasis on performance metrics has frequently overshadowed two critical aspects, resulting in a significant disparity between academic endeavors and practical application: the widespread absence of standardized, representative datasets and a uniform evaluation of thorough security evaluations. These shortcomings hinder reproducibility, obstruct equitable comparisons of authentication methods, and render systems susceptible to established biometric weaknesses.

An important point is that the problem lies in framing PPG-based authentication predominantly as a classification initiative rather than a security imperative. Achieving high accuracy in tightly controlled laboratory environments does not guarantee durability against persistent adversaries. As described in Section 5, threats such as presentation attacks, spoofing, and replay exploits pose considerable risks that are not adequately addressed in a substantial portion of the reviewed literature. Further research should implement a security-focused methodology, incorporating threat modeling, robust defenses, and extensive testing in adversarial environments. Protecting biometric templates through strategies such as cancelability, which is addressed in only a few of the research studies we reviewed, is essential to preserving user privacy and ensuring long-term system integrity.

The common reliance on modest, unique, and variably collected datasets poses a significant obstacle to progress. For the development of resilient and inclusive systems, this field demands broad, standardized datasets that capture the physiological and situational diversity of PPG signals across diverse populations and in unmonitored, everyday contexts. Emerging efforts to create publicly available datasets, as discussed in Section 6.1, mark a crucial starting point in this effort. Although new datasets represent a significant advance, tasks such as behavioral biometrics still require data collection, given their nature. Therefore, a precise specification of the hardware platforms and sensors used for collection, the techniques employed for data processing, the capture conditions, as well as the demographic and ethnic diversity is needed to provide transparent access to these data for reproducibility.

Among the promising directions for future research, prioritizing the adoption of uniform evaluation standards is crucial to ensure that emerging systems are examined not only for accuracy but also for security and operational efficiency. Designs with built-in safeguards should also be explored, as well as experiments with new multimodal strategies that go beyond conventional PPG-ECG/IMU pairings and leverage the potential of PPG-extracted behavioral biometrics. Integrating dynamic behavioral signals into ongoing multifactor authentication frameworks could represent a significant step toward secure, intuitive, and user-centric innovations.

8. Conclusions

The systematic review detailed in this research highlights a pivotal moment in the advancement of PPG-based authentication systems. Although breakthroughs in machine learning designs and multimodal fusion techniques have led to substantial gains in classification accuracy, the domain is still limited by two dominant obstacles: the scarcity

of standardized datasets and an ongoing neglect of rigorous security analyzes. These shortcomings undermine reproducibility, complicate inter-study evaluations, and expose systems to practical adversarial dangers, including spoofing and replay attacks. The dominant emphasis on refining accuracy in managed environments often overlooks the fundamental truth that biometric authentication systems should first withstand deliberate threats; a crucial reorientation is imperative.

This study offers two key contributions: first, an in-depth classification of attack vectors tailored to PPG systems and a comparative analysis including metrics for evaluation performance in authentication systems. Also, the detailed specification for sensors and platforms employed and limited availability of broad-scale, demographically inclusive datasets intensifies these issues. Recent undertakings to generate public datasets that encompass long-term and real-world variability represent an important advance, yet they require a greater level of adoption and harmonization.

In future work, fertile paths include crafting inherently secure architectures that incorporate privacy-enhancing tools such as homomorphic encryption, merging PPG with lesser-explored behavioral biometrics (for example, gesture patterns), and devising adaptive systems to counter prolonged signal fluctuations. Cross-disciplinary partnerships among signal processing specialists, cybersecurity experts, and medical professionals may reveal new insights into the physiological and environmental factors influencing PPG consistency. Additionally, research into attacks on PPG-based authentication systems in realistic environments is required to validate and evaluate the feasibility of the attacks presented in the review and those reported in various studies.

Finally, bridging the gap from lab-level precision to everyday dependability requires disciplined methodologies and a security-by-design ethos. By championing these priorities, we can foster fluid, continuous verification that harmonizes ease of use, privacy, and durability. The journey ahead is both demanding and straightforward: it is through cohesive, security-driven innovation that PPG-based authentication can realize its full potential as a foundational element of future identity verification.

CRediT authorship contribution statement

Diego Aguilar: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Alfonso Martínez-Cruz:** Writing, Validation, Supervision. **Kelsey Alejandra Ramírez-Gutiérrez:** Writing – review & editing, Validation. **Miguel Morales-Sandoval:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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